Supplementary Sensitivity Analyses for AMP-Matched Cross-Architecture Positional Robustness

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1. Seed alignment and Welch sensitivity

The ViT-B/16 and ViT-S/16 cohorts use the same six seed labels, but the architectures consume different amounts of the global random stream during model initialization. The training DataLoader does not use a separately locked generator, so equal labels do not establish identical initial weights or data order. Main-text intervals are seed-aligned difference intervals retained for continuity with the within-architecture analyses. Table S1 gives unpaired Welch intervals as a sensitivity analysis.

Table S1: Seed-aligned and unpaired Welch 95% intervals for ViT-S minus ViT-B nAUC contrasts. Both interval constructions give the same interpretive conclusion for every family and metric.

PE family	Metric	Mean difference	Seed-aligned 95% CI	Welch 95% CI
Learned	Adversarial nAUC	-0.218610	[-0.292924, -0.144295]	[-0.293806, -0.143413]
Sinusoidal	Adversarial nAUC	-0.159924	[-0.184240, -0.135608]	[-0.184361, -0.135487]
RoPE	Adversarial nAUC	-0.093204	[-0.160354, -0.026054]	[-0.151529, -0.034878]
Learned	Noise nAUC	-0.000894	[-0.003112, 0.001324]	[-0.002435, 0.000646]
Sinusoidal	Noise nAUC	0.000729	[-0.001757, 0.003216]	[-0.001043, 0.002501]
RoPE	Noise nAUC	0.000132	[-0.000479, 0.000743]	[-0.000417, 0.000681]

2. Ordinal robustness of the architecture-by-PE interaction

Let Δ_f denote ViT-S minus ViT-B adversarial nAUC for family f. More negative values indicate a larger loss under the smaller backbone. Five of the six seeds have the order Learned $<$ Sinusoidal $<$ RoPE; seed 42 is the sole exception.

Table S2: Per-seed ordering of adversarial architecture effects.

Seed	Learned Δ	Sinusoidal Δ	RoPE Δ	Ascending order
42	-0.121191	-0.159980	-0.153736	Sinusoidal < RoPE < Learned
123	-0.188993	-0.131088	-0.065368	Learned < Sinusoidal < RoPE
456	-0.200771	-0.197074	-0.157989	Learned < Sinusoidal < RoPE
789	-0.205311	-0.174361	0.014230	Learned < Sinusoidal < RoPE
1011	-0.326862	-0.149978	-0.087133	Learned < Sinusoidal < RoPE
1213	-0.268529	-0.147064	-0.109227	Learned < Sinusoidal < RoPE

The rank sums are 8, 11, and 17 for Learned, Sinusoidal, and RoPE, respectively, reproducing the exact Friedman result $Q = 7.0$, $p = 0.0289352$.

Table S3: Pairwise differences in architecture effects. Same-sign counts refer to the dominant sign across the six seeds.

Difference-in-differences contrast	95% CI	Same-sign seeds
Learned − RoPE	[-0.235330, -0.015482]	5/6 negative
Sinusoidal − RoPE	[-0.133297, -0.000143]	6/6 negative
Learned − Sinusoidal	[-0.141594, 0.024222]	5/6 negative

3. Architecture-dependent seed dispersion

Table S4: Seed dispersion of adversarial nAUC. The leave-one-out range is the minimum and maximum ViT-S/ViT-B SD ratio after omitting each seed in turn.

PE family	ViT-B range	ViT-S range	ViT-B SD	ViT-S SD	SD ratio	LOO SD-ratio range
Learned	0.007196	0.209900	0.002436	0.071663	29.414	[19.930, 47.272]
RoPE	0.035298	0.149260	0.011945	0.055722	4.665	[3.919, 5.754]
Sinusoidal	0.051913	0.059656	0.018077	0.019832	1.097	[0.614, 1.485]

The dispersion ordering, Learned $\gg$ RoPE > Sinusoidal, differs from the mean architecture-effect ordering, Learned > Sinusoidal > RoPE. This is reported descriptively; no mechanism or universal variance-scaling law is inferred.

4. Noise sign balance and normalized values above one

Table S5: Seed signs for ViT-S minus ViT-B nAUC contrasts.

PE family	Attack $\Delta < 0$	Attack $\Delta > 0$	Noise $\Delta < 0$	Noise $\Delta > 0$
Learned	6	0	3	3
Sinusoidal	6	0	2	4
RoPE	5	1	3	3

Noise contrasts have no common sign tendency, unlike the adversarial contrasts. ViT-S Sinusoidal seed 123 has noise nAUC 1.0009374. Normalized accuracy is defined relative to clean accuracy on the same finite split and is not clipped at one; a perturbed model can therefore marginally outperform its clean reference, and its integrated normalized curve can slightly exceed one.